

MOMENT OF INERTIA

Topics Covered

- Finding moment of inertia analytically and experimentally.

Prerequisites

- Hardware Interfacing laboratory experiment

1 Background

The free-body diagram of the QUBE-Servo 2 pendulum system is shown in Figure 1.1.

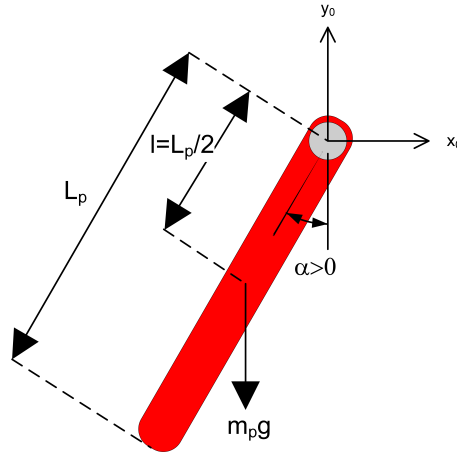


Figure 1.1: Free-body diagram of pendulum

From the free-body diagram in Figure 1.1, the resulting nonlinear equation of motion of the pendulum is

$$J_p \ddot{\alpha}(t) = m_p g l \sin(\alpha(t)), \quad (1.1)$$

where J_p is the moment of inertia of the pendulum at the pivot axis, m_p is the mass of the pendulum, L_p is the length of the pendulum (from pivot to end), and $l = L_p/2$ is the distance between the pivot and center of mass.

The moment of inertia of the pendulum can be found experimentally. Assuming the pendulum is not actuated, linearizing Equation 1.1 and solving for the differential equation yields

$$J_p = \frac{m_p g l}{(2\pi f)^2}, \quad (1.2)$$

where f is the measured frequency of the pendulum as the arm remains rigid. The frequency is calculated using

$$f = \frac{n_{cyc}}{\Delta t} \quad (1.3)$$

where n_{cyc} is the number of cycles and Δt is the duration of these cycles. Alternatively, J_p can be calculated using the moment of inertia expression

$$J = \int r^2 dm, \quad (1.4)$$

where r is the perpendicular distance between the element mass dm and the axis of rotation.

2 In-Lab Exercises

Based on the model already designed in Hardware Interfacing laboratory experiment, design a model that measures the pendulum angle using the encoder as shown in Figure 2.1.

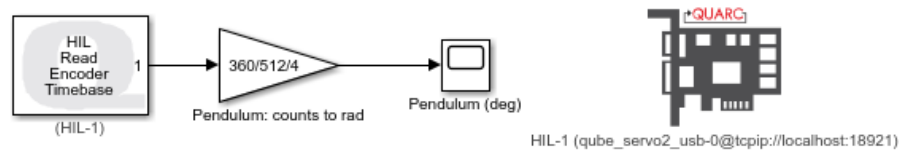


Figure 2.1: Displays measured pendulum angle

1. Find the moment of inertia acting about the pendulum pivot using the free-body diagram. Make sure you evaluate it numerically using the parameters defined in the QUBE-Servo 2 User Manual.
Hint: For solid objects with a uniform density, you can express the differential mass in terms of differential length.
2. Use the supplied **SIMULINK®** model shown in Figure 2.1 or design your own. Enter $360/(512 \cdot 4)$ in the encoder sensor gain to measure the pendulum angle in degrees.
3. Run the **QLABS®** software.
4. Load the *QUBE-Servo 2 Pendulum Virtual Experiment* by going to *QUBE 2 - Pendulum | Pendulum Workspace* virtual system.
5. In **QLABS®**, click on the **Options** button.
6. Turn ON the **Lock servo base** option and set **Pendulum lift angle (deg)** to 10 degrees. This will hold the rotary arm in place and allow you to bring the pendulum up to the 10 degree mark.
7. Run the Simulink model. With the model running, manually bring up the pendulum to the 10 degrees by clicking on the *Lift pendulum* button in **QLABS®**. The scope response should be similar to Figure 2.2.

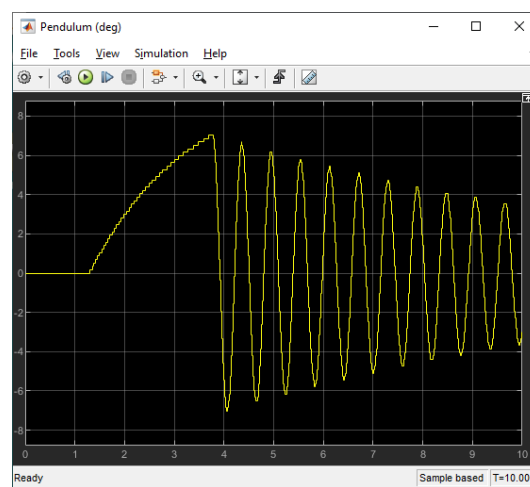


Figure 2.2: Free-oscillation response of pendulum

8. Find the frequency and moment of inertia of the pendulum using the observed results.
9. Compare the moment of inertia calculated analytically in Exercise 1 and the moment of inertia found experimentally. Is there a large discrepancy between them?

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